

CSE331 Assignment 1 Part B

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- Deadline for entire Assignment 1 Part B submission:
24 July, 2026, Friday, 11:59 pm.
 - The submitted document should have clear and legible scans.
1. Write the Regular Expressions for the following languages:
 - a. Every 1 is followed by odd number of 0's
 - b. Doesn't begin with 110
 - c. Contains substring 110 but not 00
 - d. Contains 1 in every 4th position
 - e. Doesn't contain exactly 3 0's
 - f. Has the symbol it starts with in every even position
 2. Convert the Regular Expressions to NFA:
 - a. $(0+1)^*0 + 10(0+1)^*$
 - b. $1^*(011^* + 10^*)$
 - c. $(0+01^*0)^*01^*$
 3. Convert the following DFA into Regular Expression using the State Elimination Method.
You must follow the sequence q_2, q_4, q_1, q_3
You need to show detailed steps of the process.

